

a) 160.15.0.0/19

- **Subnet Mask:** 255.255.224.0
- **Number of Subnets:** 8192 possible subnets.

1. First Subnet: 160.15.0.0 - 160.15.31.255

2. Second Subnet: 160.15.32.0 - 160.15.63.255

3. Third Subnet: 160.15.64.0 - 160.15.95.255

4. Fourth Subnet: 160.15.96.0 - 160.15.127.255

- all IPs for the fourth subnet

- **Fourth Subnet: 160.15.96.0 - 160.15.127.255**
 - **Network IP:** 160.15.96.0
 - **Router IP:** 160.15.96.1
 - **First Usable IP:** 160.15.96.2
 - **Maximum Usable IP:** 160.15.127.254
 - **Broadcast IP:** 160.15.127.255

b) 180.105.0.0/26

- **Subnet Mask:** 255.255.255.192
- **Number of Subnets:** 64 possible subnets.

- 1. First Subnet: 180.105.0.0 - 180.105.0.63**
- 2. Second Subnet: 180.105.0.64 - 180.105.0.127**
- 3. Third Subnet: 180.105.0.128 - 180.105.0.191**
- 4. Fourth Subnet: 180.105.0.192 - 180.105.0.255**

- all IPs for the fourth subnet

- Fourth Subnet: 180.105.0.192 - 180.105.0.255**
 - Network IP: 180.105.0.192**
 - Router IP: 180.105.0.193**
 - First Usable IP: 180.105.0.193**
 - Maximum Usable IP: 180.105.0.254**
 - Broadcast IP: 180.105.0.255**

c) 240.108.90.0/28

- Subnet Mask: 255.255.255.240**
- Number of Subnets: 16 possible subnets.**

- 1. First Subnet: 240.108.90.0 - 240.108.90.15**
- 2. Second Subnet: 240.108.90.16 - 240.108.90.31**
- 3. Third Subnet: 240.108.90.32 - 240.108.90.47**
- 4. Fourth Subnet: 240.108.90.48 - 240.108.90.63**

- all IPs for the fourth subnet

- **Fourth Subnet: 240.108.90.48 - 240.108.90.63**
 - **Network IP: 240.108.90.48**
 - **Router IP: 240.108.90.49**
 - **First Usable IP: 240.108.90.49**
 - **Maximum Usable IP: 240.108.90.62**
 - **Broadcast IP: 240.108.90.63**

d) 105.0.0.0/21

- **Subnet Mask: 255.255.248.0**
- **Number of Subnets: 2048 possible subnets.**

- 1. First Subnet: 240.108.90.0 - 240.108.90.15**
- 2. Second Subnet: 240.108.90.16 - 240.108.90.31**
- 3. Third Subnet: 240.108.90.32 - 240.108.90.47**
- 4. Fourth Subnet: 240.108.90.48 - 240.108.90.63**

- all IPs for the fourth subnet

- **Fourth Subnet: 105.0.24.0 - 105.0.31.255**
 - **Network IP: 105.0.24.0**
 - **Router IP: 105.0.24.1**
 - **First Usable IP: 105.0.24.1**
 - **Maximum Usable IP: 105.0.31.254**
 - **Broadcast IP: 105.0.31.255**